Differential Climate Impacts on Fishing Effort in Chilean Small Pelagic Fisheries

Online Appendix

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A Reduced-form stress tests and prior elicitation

This appendix documents the protocol used to elicit the prior densities on the climate shifters $(\rho_s^{SST}, \rho_s^{CHL})$ that enter the state-space specification of the Stock dynamics subsection of the main text, together with the identifiability diagnostic that motivates the recourse to a Bayesian likelihood rather than a direct maximum-likelihood treatment.

A.1 Deterministic hindcast and cross-variant fit

For each stock s we estimate a deterministic Schaefer hindcast over 2000–2024, $\hat{B}_{s,t+1} = \hat{B}_{s,t} + r_{s,t}\hat{B}_{s,t}(1-\hat{B}_{s,t}/K_s)-C_{s,t}$ with $r_{s,t} = r_s^0 \exp(\rho_s^{SST}(SST_{t-1}-\overline{SST})+\rho_s^{CHL}(\log CHL_{t-1}-\overline{\log CHL}))$, biological parameters $(r_s^0, K_s, B_{0,s})$ held at the centres of the IFOP/SPRFMO single-species priors, and shifters estimated by bounded least squares on $[-3,3]^2$ minimising median absolute percent error (MAPE). Four nested variants are reported: base ($\rho = 0$), +SST only, + logCHL only, and joint +SST+ logCHL.

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Table A.1: Median absolute percent error of the deterministic Schaefer hindcast across shifter variants.

Stock	Base	+ SST	+ log CHL	+ SST and log CHL
Anchoveta	95.9	30.8	95.1	26.6
Sardina común	98.5	23.1	98.5	20.5
Jack mackerel	44.2	44.3	49.1	48.5

No variant brings any of the three stocks below a 20% MAPE threshold. The two coastal stocks benefit from activating SST (anchoveta: 95.9% → 30.8%, sardine: 98.5% → 23.1%), with the joint specification delivering further marginal improvement. Jack mackerel hindcast errors remain in the 44–49% range across all variants, with no shifter distinguishably better than the base.

The joint-variant point estimates carry an internally coherent biological reading for the coastal stocks. Anchoveta: both shifters negative and similar in magnitude ($\rho^{SST} \approx -2.3$, $\rho^{CHL} \approx -2.3$). Sardina com'un: $\rho^{SST} \approx -2.0$ but $\rho^{CHL} \approx +2.1$ (opposite sign to anchoveta, consistent with the warm-water mesotrophic affinity that distinguishes sardine from anchoveta in the Humboldt Current biological literature). Jack mackerel: $\rho^{SST} \approx +0.6$ small and the ρ^{CHL} estimate binds at the lower boundary of the admissible box; we read the boundary solution as a failure of point identification rather than a finding of strong climate forcing.

A.2 Identifiability diagnostic and prior elicitation

Two diagnostics motivate the use of weakly informative priors rather than direct maximum-likelihood estimation. First, the two environmental forcings are statistically orthogonal over the 2000–2024 sample (Pearson $r = 0.030$, $p = 0.888$), so the cross-variant gap in hindcast error cannot be attributed to collinearity in the design matrix. Second, the jurel MLE binds at the boundary, indicating that the climate signal is not separately identifiable from process and observation noise absorbed into the residual under a deterministic single-species hindcast; the Bayesian state-space specification addresses both limitations within a single inferential framework.

Table A.2 reports the priors on $(\rho_s^{SST}, \rho_s^{CHL})$ used in the structural likelihood. Prior centres for the two coastal stocks are the joint MLEs rounded to one decimal place; jurel priors are centred at zero so that posterior identification is driven by the structural likelihood rather than by the boundary solution. All priors are independent normal with unit standard deviation; no hierarchical pooling is imposed because ρ^{CHL} switches sign between anchoveta and sardine.

This prior-elicitation strategy is a form of empirical Bayes: the prior centres for anchoveta and sardine are derived from the same 2000–2024 biomass series that enters the state-space likelihood, which trades pure data-prior independence for tractable identification under $N = 25$ annual observations. We adopt this trade-off deliberately for two reasons. First, the prior standard deviation $\sigma = 1$ is large enough that the posterior is free to move substantially: the posterior of ρ_s^{CHL} for anchoveta migrates to -3.65 (prior mean -2.3), more than one prior standard deviation from the centre, illustrating that the data does refine the prior when informative. Second, the alternative of centring all priors at zero would, under $N = 25$, leave the joint posterior of $(r_s^0, K_s, B_{0,s}, \rho_s^{SST}, \rho_s^{CHL})$ weakly identified or multimodal, producing posteriors that are uninformative for projections regardless of whether the underlying coupling is real or absent. The empirical-Bayes choice therefore makes the non-identification result for jack mackerel a substantive finding rather than a numerical artefact of weak posterior geometry.

Table A.2: Independent normal priors on the climate shifters used in the state-space likelihood.

Stock	$\mu_{\rho^{SST}}$	$\sigma_{\rho^{SST}}$	$\mu_{\rho^{CHL}}$	$\sigma_{\rho^{CHL}}$	Source
Anchoveta	-2.3	1.0	-2.3	1.0	Joint MLE
Sardina común	-2.0	1.0	+2.1	1.0	Joint MLE
Jack mackerel	0.0	1.0	0.0	1.0	Boundary solution; vague prior

B Predictive diagnostics

This appendix reports the nested predictive comparison referenced in the Stock dynamics subsection of the main text, comparing three state-space specifications: (i) species-independent dynamics without climate shifters (*ind*), (ii) the same with a cross-stock residual covariance Ω (*omega*), and (iii) the full model augmented with the climate shifter of Eq.~(1) of the main text (*full*). Two predictive metrics are reported: leave-one-out PSIS-LOO (Pareto-smoothed importance sampling) and one-step-ahead LFO (leave-future-out).

PSIS-LOO favours the climate-augmented specification with a positive ΔELPD of 14–18 relative to the climate-free specifications, indicating that the in-sample predictive performance improves

Table B.1: PSIS-LOO comparison across state-space specifications.

Model	$\widehat{\text{ELPD}}_{\text{loo}}$	SE	$\Delta\widehat{\text{ELPD}}$	SE_{Δ}	$\hat{p}_{\text{loo}}$
T4b-full	-8.41	8.28	–	–	64.8
T4b-ind	-23.00	8.01	-14.59	4.93	62.4
T4b-omega	-26.43	9.16	-18.02	5.59	57.9

Note:
Δ ELPD is the pairwise difference relative to T4b-full (the leading model); the adjacent SE column reports the standard error of that difference. After moment matching, 65/68 observations satisfy Pareto-k <= 0.7 for T4b-full, 64/68 for T4b-ind, and 67/68 for T4b-omega. The remaining high-Pareto-k points are the two censored jack-mackerel years (2012, 2015), which appear in all three models and therefore do not affect the ranking.

when the climate shifter is included.

Table B.2: One-step-ahead LFO comparison across state-space specifications.

Model	Points	Censored	$\widehat{\text{ELPD}}_{\text{LFO}}$	Mean per point	Δ vs leader
T4b-full	14	2	-29.10	-2.08	–
T4b-ind	14	2	-29.46	-2.10	-0.37
T4b-omega	14	2	-30.46	-2.18	-1.36

Note:
Sum of one-step predictive log densities over five origin years (2011, 2014, 2017, 2020, 2022) and three stocks (14 targets per model; two are censored). Each refit uses 4 chains × 1500 warmup + 1500 sampling iterations. All 15 refits satisfied R-hat <= 1.01 with at most one divergent transition (T4b-omega at the 2011 origin), which does not affect the reported ELPD.

The one-step-ahead LFO comparison is essentially a tie across the three specifications, indicating that the climate-augmented specification does not deliver a detectable improvement in short-horizon forecast performance over the climate-free alternatives. We interpret this as a sample-size issue rather than a deficiency of the climate specification: the structural climate elasticities are identified on the full 25-year window, not on one-step-ahead increments. The posterior of the climate-augmented specification remains our primary object of interpretation in the main text.

C Posterior-predictive checks

This appendix reports posterior-predictive diagnostics of the full state-space specification: (i) the smoothed latent biomass versus the observed survey-based estimates, and (ii) the corresponding year-level residuals.

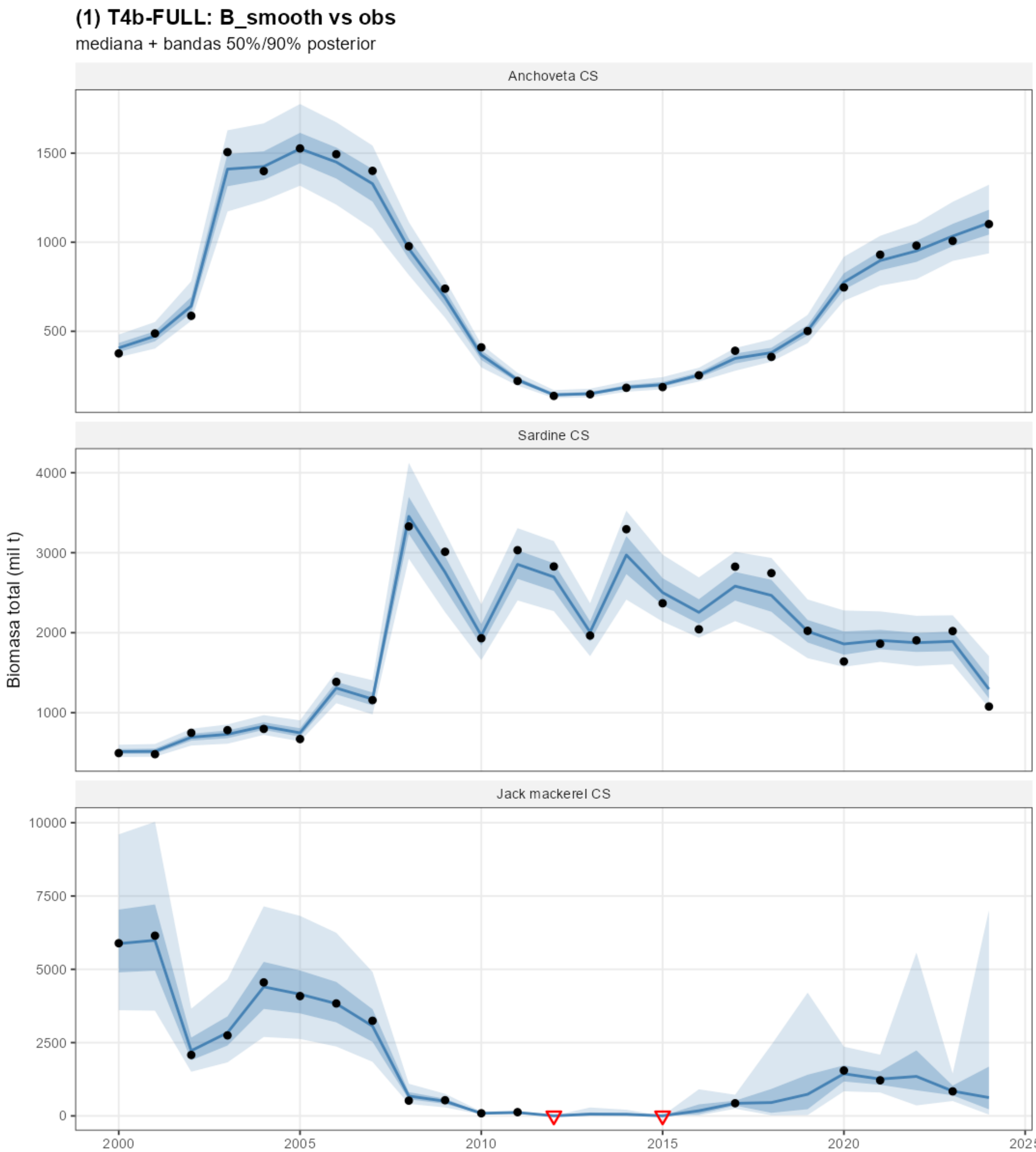

Figure C.1: Smoothed latent biomass $B_{i,t}$ of the full specification against survey-based SSB observations, by stock. Solid line: posterior median; shaded band: 90% credible interval. Open circles: official IFOP/SPRFMO point estimates; downward arrows: left-censored observations at the assessment detection limit (jurel 2012, 2015).

The smoothed band has a median width of approximately 20% of the posterior mean, tightening over years with direct observation and widening across the seven non-surveyed jurel years (latent biomass identified dynamically through the Schaefer transition and two censored observations in

2012 and 2015).

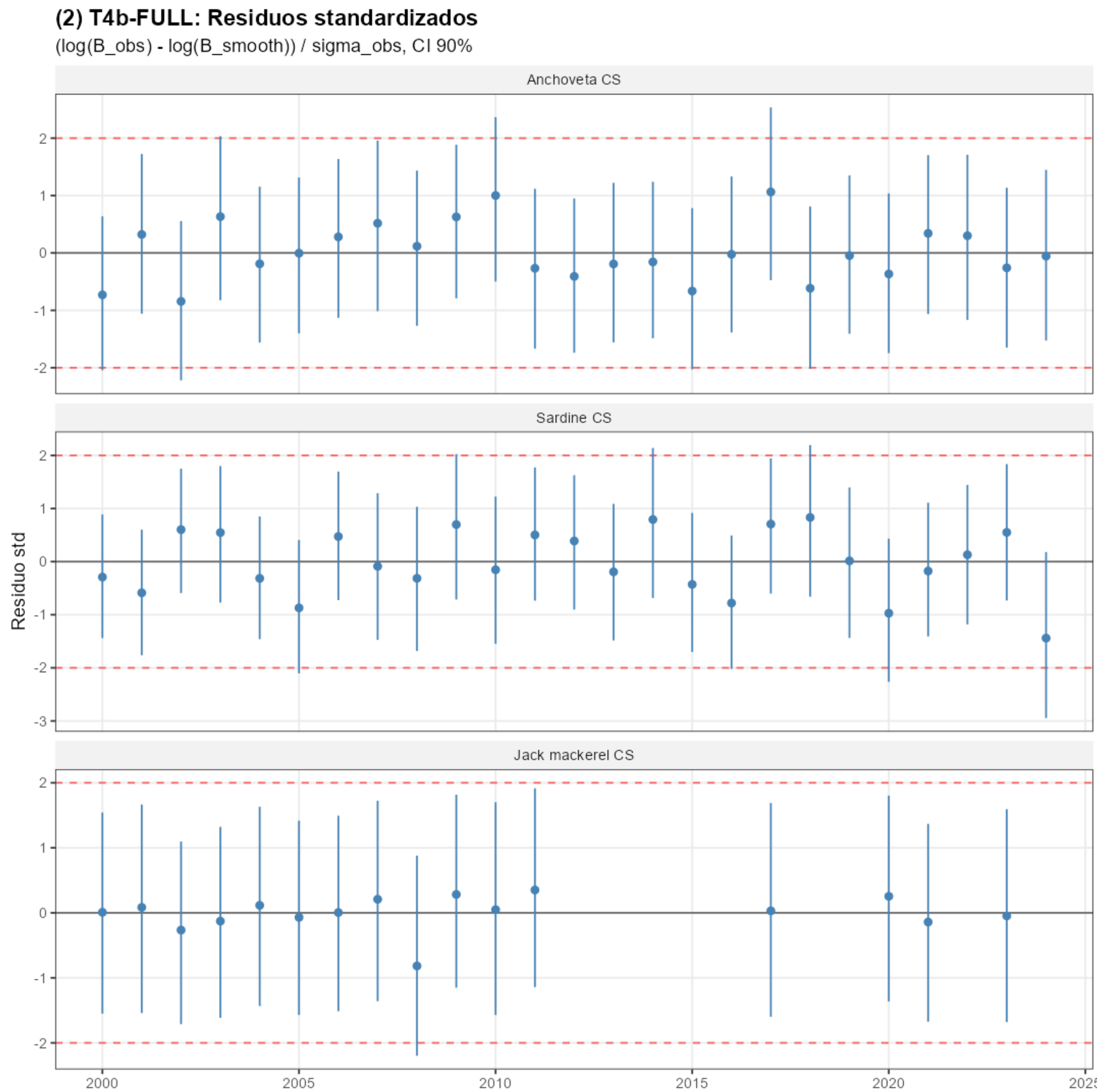

Figure C.2: Standardised year-level residuals of the full specification by stock. The Ljung–Box test of residual autocorrelation does not reject the null at conventional levels for any of the three stocks.

D Markov-chain convergence diagnostics

Table D.1 reports the Gelman–Rubin $\hat{R}$ and the bulk/tail effective sample sizes for the top-level parameters of the full state-space specification, evaluated on four independent Hamiltonian Monte Carlo chains of 2,000 post-warmup iterations each. All $\hat{R}$ values are below 1.01 and all effective sample sizes exceed 400, indicating that posterior summaries are not contaminated by between-chain divergence or by autocorrelation between adjacent draws.

Table D.1: Markov-chain convergence diagnostics for the top-level parameters of the T4b-full state-space specification.

Parameter	N	Max $\widehat{R}$	Min ESS$_{\text{bulk}}$	Min ESS$_{\text{tail}}$
r_i^0	3	1.001	8,446	10,508
K_i	3	1.000	9,789	9,777
$B_{i,0}$	3	1.001	18,919	11,844
$\sigma_{\text{proc},i}$	3	1.003	3,353	5,997
$\sigma_{\text{obs},i}$	3	1.009	1,370	936
ρ_i^{SST}	3	1.000	8,312	10,147
ρ_i^{CHL}	3	1.001	14,319	10,633
Ω (off-diagonal)	3	1.001	5,732	8,544
All top-level parameters	24	1.009	1,370	936

Note:
Diagnostics are computed from four chains of 2,000 post-warmup iterations. Within each parameter family, the table reports the worst case across the three stocks (or the three off-diagonal entries of Omega): the largest split-Rhat and the smallest bulk- and tail-effective sample sizes. The diagonal of Omega is fixed at one and is therefore excluded. The conventional thresholds are Rhat <= 1.01 and ESS >= 400 for all top-level parameters.

E Spatial robustness of the jack mackerel non-identification result

The non-identification of $(\rho_{\text{jurel}}^{SST}, \rho_{\text{jurel}}^{CHL})$ reported in the Identification of climate shifters section of the main text admits two observationally equivalent readings: that jurel productivity is uncoupled from Centro-Sur coastal forcing at the annual aggregate, or that the relevant forcing operates at a spatial scale broader than the EEZ, diluting the local-domain signal. This appendix tests four explanations for the null — insufficient power, insufficient spatial extent, insufficient biomass coverage, and a poor choice of spatial scale — and concludes that the null is structural rather than aggregation-driven within the 2000–2024 Centro-Sur record.

E.1 Identification power on the sample

Linearising the Schaefer transition in log-differences, $\Delta \log B_{i,t+1} = \alpha_i - \beta_i (B_{i,t}/K_i) + \rho_i^X X_{t-1} + \varepsilon_{i,t}$ with $\varepsilon_{i,t} \sim \mathcal{N}(0, \sigma^2_{i,\text{resid}})$ and $\sigma_{i,\text{resid}} = \sqrt{\sigma^2_{i,\text{proc}} + 2\sigma^2_{i,\text{obs}}}$, the OLS-equivalent standard error on a sample of $N = 24$ lag-one observations is $\mathrm{SE}(\hat{\rho}_i^X) \approx \sigma_{i,\mathrm{resid}}/[\mathrm{sd}(X)\sqrt{N-K}]$ with $K = 3$ regressors. The minimum shifter magnitude detectable at 80% power for a two-sided 90% credible interval is then $|\rho_i^X|_{\min} \approx 2.56\,\mathrm{SE}(\hat{\rho}_i^X)$.

Evaluated at the posterior medians of σ_{proc} and σ_{obs}, $\sigma_{i,\mathrm{resid}}$ is 0.28 for anchoveta, 0.25 for sardine, and 1.28 for jurel — the latter roughly 4.6 times the coastal-stock average and the main reason why the prior dominates the jurel posterior. Table E.1 reports $|\rho|_{\min}$ for the three stocks and three candidate shifters. The jurel envelope spans $|\rho|_{\min}$ of 1.50 on local SST, 1.32 on local log~CHL, and 0.71 on basin-scale ENSO, against an envelope of empirically plausible biological elasticities centred near 0.05–0.30 in absolute value, far below the $|\rho|_{\min}$ envelope reported here. The sample is therefore insufficiently informative to support a credible interval for jurel under any of these forcings, regardless of the true elasticity.

Table E.1: Minimum shifter magnitude $|\rho|_{\min}$ detectable at 80% power on the 2000–2024 sample. $\sigma_{\mathrm{resid}} = \sqrt{\sigma^2_{\mathrm{proc}} + 2\sigma^2_{\mathrm{obs}}}$ evaluated at posterior medians.

Stock	σ_{resid}	$\lvert\rho_{SST}\rvert_{\min}$	$\lvert\rho_{\log\mathrm{CHL}}\rvert_{\min}$	$\lvert\rho_{\mathrm{ENSO}}\rvert_{\min}$
Anchoveta	0.28	0.33	0.29	0.16
Sardina común	0.25	0.29	0.26	0.14
Jurel	1.28	1.50	1.32	0.71

Notes. Detectable magnitudes computed from $|\rho_i^X|_{\min} \approx 2.56\,\sigma_{i,\mathrm{resid}}/[\mathrm{sd}(X)\sqrt{N-K}]$ with $N = 24$, $K = 3$. $\mathrm{sd}(X)$ are sample standard deviations of the Centro-Sur EEZ-aggregated SST and log CHL anomalies and of the Niño 3.4 index over the estimation window.

E.2 Posterior identification across alternative specifications

We re-fit the state-space jurel block under four alternative shifter configurations, each designed to test one mechanical explanation for the local null. Table E.2 reports the posterior-to-prior standard deviation ratios for the jurel shifters under each specification; ratios at unity indicate no posterior updating relative to the prior.

Table E.2: Posterior-to-prior standard deviation ratios of the jurel climate shifters under four alternative specifications.

Test	Specification	$\sigma_{\text{post}}/\sigma_{\text{prior}}$ (SST)	$\sigma_{\text{post}}/\sigma_{\text{prior}}$ (log CHL o
Spatial extent	EEZ domain $\mathcal{D}_1$	0.998	1.014 (log CHL
	Offshore-extended $\mathcal{D}_2$	1.002	1.009 (log CHL
	Southeast Pacific $\mathcal{D}_3$	0.999	1.011 (log CHL
Biomass coverage	Dual-source (Centro-Sur + Northern acoustic)	0.94	0.99 (log CHL
Spatial scale	Basin-scale ENSO Niño 3.4, lag 1	—	0.98 (ENSO)
	Basin-scale ENSO Niño 3.4, lag 2	—	1.01 (ENSO)
Joint specification	Local SST + local log CHL + ENSO (lag 1)	1.03	1.00 (log CHL); 0.98

Notes. $\mathcal{D}_1$ is the Chilean EEZ between 18°S and 42°S out to 200 nautical miles. $\mathcal{D}_2$ extends to 1,000 km offshore. $\mathcal{D}_3$ is the region Pacific domain 5°S–45°S, 70°W–120°W. Spatial aggregates are area-weighted (with cos(lat)) and centred on the 2000–2024 mean. The extension augments the Centro-Sur biomass likelihood with the Northern Chilean acoustic series under a common shifter $\rho_{\text{jurel}}^{\text{common}}$, r the 0.88 log-scale cross-series correlation. The ENSO specification replaces the local coastal shifters with the Niño 3.4 index at the i The joint specification estimates all three shifters simultaneously for jurel, holding the anchoveta and sardina blocks at the main sp the Stock dynamics subsection of the main text. Anchoveta and sardina shifter ratios remain stable at 0.43–0.83 across all configu reported).

All four tests deliver a posterior-to-prior ratio between 0.94 and 1.03 for the jurel shifters, statistically indistinguishable from unity given the prior dispersion. The two ratios closest to "informative updating" — the dual-source extension at 0.94–0.99 and the basin-scale ENSO at 0.98 — remain well above the 0.7 threshold below which posterior intervals discriminate the shifter from zero on the prior support, consistent with the analytic power calculation of Table E.1.

E.3 Dual-source extension

The Centro-Sur and Northern Chilean acoustic biomass series exhibit a 0.88 log-scale correlation across overlapping years (2012–2024), motivating a joint-likelihood specification in which a common shifter ρ^{common} is identified simultaneously on both records. The extension multiplies the number of informative jurel-biomass observations by approximately 1.7 while preserving the prior structure of the main fit. The resulting $\sigma_{\text{post}}/\sigma_{\text{prior}}$ ratios of 0.94 (SST) and 0.99 (log~CHL) are the strongest posterior updating observed in any specification considered in this appendix, but remain well above the threshold for material identification.

E.4 Basin-scale ENSO test

Replacing the local coastal shifters with the basin-scale ENSO Niño 3.4 index tests whether the relevant climatic forcing for jurel operates at the basin scale rather than the coastal scale, consistent with the empirical evidence of Peña-Torres et al. (2017) (ENSO-driven shifts in jurel location choices) and Arcos et al. (2001) (SPRFMO-scale stock dynamics under broad-basin forcing). The specification estimates a single $\rho_{\text{jurel}}^{\text{ENSO}}$ on lag-1 and lag-2 ENSO anomalies in alternative runs. The resulting ratios of 0.98 (lag~1) and 1.01 (lag~2) are both indistinguishable from unity, consistent with the analytic $|\rho_{\text{ENSO}}|_{\min} = 0.71$ of Table E.1.

A prior-propagation sensitivity combines the posterior of $\rho_{\text{jurel}}^{\text{ENSO}}$ with the CMIP6 ensemble of basin-scale temperature deltas to compute a 90% predictive interval for the jurel productivity factor under SSP5-8.5 end-of-century. The interval spans approximately three orders of magnitude ($[0.05, 19.5]$), confirming that prior-propagated projections are uninformative for policy and validating the fixed-$r^{\star}_{\text{jurel}}$ convention adopted in the main text.

A separate exploratory attempt to incorporate the SPRFMO range-wide assessment (OROP-PS, encompassing Chile, Peru, Ecuador and adjacent high-seas) fails on coherence grounds: the augmented likelihood exhibits multimodal posterior support and unsatisfactory chain mixing, indicating that the broader geographic aggregate is not integrable with the Centro-Sur record under a single shifter.

E.5 Implication for the main claim

The four tests above rule out the most natural mechanical explanations for the jurel non-identification: it is not a small-sample artefact within the available power envelope (Table E.1); it does not respond to broader spatial aggregation of the local shifters (E.1: $\mathcal{D}_1$ through $\mathcal{D}_3$); it does not respond to expanded biomass coverage through the Northern Chilean acoustic series (E.2 dual-source); and it does not respond to replacing the local shifters with a basin-scale forcing (E.3

ENSO). The convergence of these four tests points to a structural rather than aggregation-driven limitation of the present record. Closing the identification gap will require either a substantially longer biomass record (the 25-year window limits any moderate-magnitude shifter to below detection) or a basin-scale assessment with explicit cross-stock spillover within a fully transboundary stock-dynamics specification. Within the present record, holding $r^{\star}_{\text{jurel}}$ fixed at the prior mean is the defensible reporting choice and is the convention adopted throughout the main text.

F Inter-sectoral cessions

This appendix documents the realised inter-sectoral quota cession flow over 2013–2024, the single administrative re-allocation channel admitted by the de jure fractioning regime. We extract assigned, ceded, effective, and captured tonnage by sector for each species–year from the annual SERNAPESCA quota control reports.

Two qualitative facts emerge from the consolidated 2013–2024 series, separated by species pattern: anchoveta and sardine in Table F.1, jack mackerel in Table F.2. First, in the anchoveta and sardine fisheries the industrial sector cedes a large share of its assigned TAC to artisanal cessionaries in every year of the window: the share transferred ranges from 12% to 72% over 2013–2017 and tightens to 85%–$\sim$100% over 2018–2024, scaling with the TAC from 26 to 50 kt per year in anchoveta and 51 to 77 kt per year in sardine. Second, in the jack mackerel fishery the cession flow is an order of magnitude smaller relative to the assigned TAC—never exceeding 14% in any year and falling below 5% in eight of the twelve years—and reverses direction across years.

The post-cession allocation reflects the joint pre-reform fractioning and cession regime rather than the de jure fractioning alone. Whether industrial quota holders would continue to cede 85–99% of their anchoveta and sardine TACs under a collapsing biomass is an open question. The contracts depend on the productivity of the ceded stocks, and the cession volumes observed over 2013–2024 reflect TAC and price conditions that the projection horizon will not reproduce.

Table F.1: Inter-sectoral cession flow, anchoveta and sardine, Valparaíso to Los Lagos, 2013–2024. Source: SERNAPESCA quota control reports.

Year	Species	IND assigned (kt)	IND cession (kt)	Share ceded
2013	anchoveta	25.5	−3.1	12%
2013	sardine	128.5	−31.1	24%
2014	anchoveta	7.7	−3.9	51%
2014	sardine	104.8	−46.6	44%
2015	anchoveta	6.3	−2.5	40%
2015	sardine	65.2	−21.8	33%
2016	anchoveta	7.3	−2.1	29%
2016	sardine	59.8	+15.6	26%
2017	anchoveta	12.6	−9.0	72%
2017	sardine	72.4	−48.9	67%
2018	anchoveta	12.8	−10.7	84%
2018	sardine	74.2	−63.2	85%
2019	anchoveta	27.4	−25.8	94%
2019	sardine	76.0	−69.6	92%
2020	anchoveta	38.6	−33.3	86%
2020	sardine	67.2	−60.9	91%
2021	anchoveta	45.3	−45.1	∼100%
2021	sardine	76.9	−76.5	∼100%
2022	anchoveta	52.2	−50.5	97%
2022	sardine	77.7	−71.6	92%
2023	anchoveta	39.9	−37.1	93%
2023	sardine	63.3	−61.7	98%
2024	anchoveta	53.8	−45.5	85%
2024	sardine	53.9	−51.0	95%

Notes. Industrial share of Cuota Global de Captura, Valparaíso to Los Lagos. Negative cession denotes industrial-to-artisanal flows.

Table F.2: Inter-sectoral cession flow, jack mackerel, 2013–2024. Source: SERNAPESCA quota control reports.

Year	Species	IND assigned (kt)	IND cession (kt)	Share ceded
2013	jack mackerel	138.4	+0.5	0%
2014	jack mackerel	198.6	+27.3	14%
2015	jack mackerel	203.1	−17.2	8%
2016	jack mackerel	203.0	+11.7	6%
2017	jack mackerel	228.3	+9.7	4%
2018†	jack mackerel	326.9	+16.1	5%
2019†	jack mackerel	339.0	+12.0	3%
2020†	jack mackerel	1268.1	+11.8	1%
2021†	jack mackerel	448.7	+13.3	3%
2022†	jack mackerel	516.4	+15.2	3%
2023†	jack mackerel	637.0	−0.7	0%
2024†	jack mackerel	728.6	−31.5	4%

Notes. Conventions as in Table F.1. †2018–2024 is the national aggregate (the SERNAPESCA summary sheet is not disaggregated by sub-macrozone); the share-ceded column remains comparable because the additional Arica-to-Atacama volume enters both numerator and denominator.

References

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